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BayesReconPy: A Python package for forecast reconciliation

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Software

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Summary

BayesReconpy implements methods for probabilistic forecast reconciliation in Python. It reconciles hierarchies containing real-valued time series, discrete time series and a mixture of real-valued and discrete time series (mixed hierarchies). The package is released under the LGPL (3) license and available on GitHub.

Statement of Need

Forecast reconciliation is essential for ensuring coherence across hierarchical time series, where aggregate values must match the sum of their components. A common example is retail sales: item-level sales can be aggregated by product, category, or store (Makridakis et al., 2022). While the observed data are coherent by construction, independently generated forecasts (base forecasts) typically violate aggregation constraints, resulting in incoherent predictions.

Early work addressed this issue for point forecasts using projection-based methods such as OLS and MinT (Rob J. Hyndman et al., 2011; Wickramasuriya et al., 2019). More recently, probabilistic reconciliation methods have been proposed to reconcile entire forecast distributions (Girolimetto et al., 2024; Jeon et al., 2019; Panagiotelis et al., 2023; Taieb et al., 2021), offering a richer and more informative forecasting framework.

Despite this progress, most existing software tools for reconciliation have significant limitations (**Table 1**). Many only support Gaussian or continuous distributions, lack support for discrete or mixed-type forecasts, or are implemented only in R. Others, such as ProbReco (Panagiotelis et al., 2023) and DiscreteRecon (Zhang et al., 2024), are no longer actively maintained. Notably, no existing Python library offers a unified interface for probabilistic reconciliation across distribution types.

bayesReconPy fills this gap. It is a Python package that implements probabilistic reconciliation via both conditioning and projection-based methods, including OLS and MinT. It supports a broad range of forecast types:

- Gaussian forecasts (Corani et al., 2021)
- Continuous non-Gaussian forecasts (Zambon, Azzimonti, & Corani, 2024)
- Discrete forecasts (Corani et al., 2024; Zambon, Azzimonti, & Corani, 2024)
- Mixed discrete-continuous hierarchies (Zambon, Azzimonti, Rubattu, et al., 2024)

bayesReconPy is an extension of the R package bayesRecon (Azzimonti et al., 2024), offering the same functionality in a Python-native interface. As shown in a comparison with other reconciliation libraries, it is the only actively maintained Python package that enables probabilistic

reconciliation across all major distribution types. It includes comprehensive documentation and Python notebooks that replicate key experiments from the literature (Corani et al., 2021, 2024; Zambon, Azzimonti, & Corani, 2024), making it accessible for researchers and practitioners alike.

Table 1: Probabilistic reconciliation methods comparison

Library	Cross-temp	Gauss-ian	Continuous (non-Gaussian)	Dis-crete	Mixed
bayesReconPy (Ours)	X	V	V	V	V
fable / fabletools (O'Hara-Wild et al., 2024)	V	V	V	X	X
FoReco (Girolimetto & Fonzo, 2024)	V	V	V	X	X
gluonts (Alexandrov et al., 2020)	X	V	V	X	X
hierarchicalforecast (Olivares et al., 2022)	X	V	V	X	X
thief (Rob J. Hyndman & Kourentzes, 2018)	V	V	X	X	X

Note: V = Supported, X = Not supported

Usage

A hierarchy can contain, at each level, Gaussian, continuous non-Gaussian, or discrete forecast distributions (see **Figure 1**). Continuous base forecasts can be provided either in parametric form or as samples. Discrete base forecasts can be provided in parametric form, as samples, or as a probability mass function (PMF).

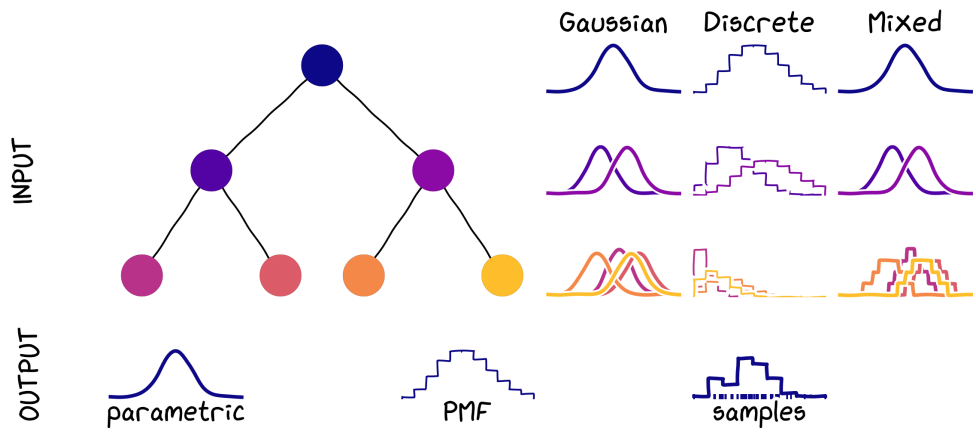

Figure 1: Types of reconciliation — Gaussian, discrete, and mixed input forecasts, and output forms (parametric, PMF, samples).

Below, we describe the suitable reconciliation algorithms for each case:

1. **All forecast distributions are Gaussian**
   Reconciliation is computed analytically using the function `reconc_gaussian`, following the approach in (Corani et al., 2021; Zambon, Agosto, et al., 2024). This is equivalent to minimum trace reconciliation (MinT) (Wickramasuriya et al., 2019).

2. **All forecast distributions are continuous (non-Gaussian) or all are discrete**
   Reconciliation is performed using Bottom-Up Importance Sampling (BUIS), via the function `reconc_BUIS` (Zambon, Azzimonti, & Corani, 2024).

3. **Mixed distributions: discrete forecasts at the bottom level and Gaussian at the upper levels**
   These hierarchies can be reconciled using either Mixed Conditioning (`reconc_MixCond`) or Top-Down Conditioning (`reconc_TDcond`) (Zambon, Azzimonti, Rubattu, et al., 2024).

Depending on the reconciliation function used, the output is returned as:

- *Reconciled parameters* (for the Gaussian case, using `reconc_gaussian`)
- *Samples* (for continuous non-Gaussian or discrete forecasts, using `reconc_BUIS`)
- *Probability Mass Functions (PMFs)* (for mixed cases, using `reconc_MixCond` or `reconc_TDcond`)

Note that in the case of MinT or OLS reconciliation, the input is expected to be numpy arrays and the same is returned as the reconciled forecast. Documentation for the expected shape of these arrays is provided in the function descriptions.

Examples

Reconciliation of Negative Binomial Forecasts

We demonstrate the use of `bayesReconPy` on a hierarchy of extreme market events in five economic sectors over the period 2005–2018. This hierarchy consists of five bottom-level and one top-level time series (the total). These are count-valued time series, and the predictive distributions are modeled as negative binomial.

The dataset `extr_mkt_events`, included in the package, contains both the observed time series and the corresponding base forecasts. It was used in the experiments of (Zambon, Agosto, et al., 2024). A related Python notebook reproducing the results is available: Properties of the Reconciled Distribution via Conditioning

The code below shows how to apply the `reconc_BUIS` function. The function takes the summing matrix `A`, the base forecast parameters, and the desired number of samples. Reconciliation is completed within seconds:

```
# Reconcile via importance sampling
buis = reconc_BUIS(A, base_fc_j,
        "params", "nbinom",
        num_samples=N_samples, seed=42)
samples_y = buis['reconciled_samples']
# Computational time for 3508 reconciliations: 20.13 seconds
```

Reconciliation of a Large Mixed Hierarchy

The M5 forecasting competition dataset (Makridakis et al., 2022) includes daily sales time series for 10 stores. Each store contains 3049 bottom-level series and 11 upper-level series.

During the competition, existing reconciliation methods failed to process the hierarchy due to the dataset's size and the requirement for non-negative forecasts (Makridakis et al., 2022). `bayesReconPy` successfully reconciles 1-step-ahead base forecasts for one store, `"CA_1"`, returning non-negative and probabilistic forecasts.

Base forecasts are included in the package and were generated using the ADAM method (Svetunkov & Boylan, 2023), implemented in the `smooth` R package (Svetunkov, 2024). A Python notebook illustrates this example, available as: Reconciliation of M5 hierarchy with mixed-type forecasts

Below is a code snippet demonstrating reconciliation using `reconc_TDcond`. Here, the bottom-level forecasts are discrete, and the upper-level forecasts are continuous. The reconciliation

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completes in just a few seconds:

```
N_samples_TD = int(1e4)

# TDCond reconciliation
start = time.time()
td = reconc_TDcond(
    A,
    fc_bottom_4rec,
    fc_upper_4rec,
    bottom_in_type="pmf",
    num_samples=N_samples_TD,
    return_type="pmf",
    seed=seed
)
stop = time.time()

# Reconciliation time: 10.11 s
```


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